

Results Evidence Map for *The Making of Vehicle Currencies*

Internal results packet. Not manuscript prose.

Purpose and Reading Order

This document reorganizes the current empirical evidence around research questions, not around the order in which the analyses were built. Table 1 is the master map: every empirical claim must point to one research question, one proxy, and one primary table. The tables use a JFE-style hierarchy: compact headline estimates in the main rows, conventional significance stars shown on estimates, p-values shown explicitly, fixed effects and clustering reported in notes, and weaker or conditional evidence labelled as such.

Interpretation Rules

1. Definitions are not propositions. Vehicle use is defined by intermediate-token use in in-direct routes; the propositions are about emergence, persistence, rotation, architecture, and settlement design.
2. “Value” is not a primitive. Feasibility, depth/capacity, execution-cost advantage, settlement-transfer incidence, and LP-liquidity response are the measurable objects.
3. WETH, stablecoins, V3, and V4 are empirical test beds. They should not be written as the propositions themselves.
4. Conventional significance stars are shown on estimates (*, **, *** for 10%, 5%, and 1%). P-values are also reported directly so significance is not hidden behind stars. Weak, noisy, or pre-trending results are labelled in the table notes.

Table 1: Research questions, empirical objects, and evidence.

RQ	Research question	Unit	Main proxy	Evidence	Empirical answer
RQ1	When does one asset become the vehicle?	Endpoint pair x candidate vehicle x day.	Route-cost advantage, vehicle-route availability, vehicle depth.	Table 4	Actual vehicle share and future vehicle share rise with executable vehicle-route economics.
RQ2	How does liquidity provision make a vehicle?	Token x day dynamic panel.	LP concentration, vehicle-linked liquidity, lagged vehicle share.	Table 5	Vehicle use and vehicle-linked liquidity are mutually persistent in stock allocations.
RQ3	Why does vehicle status persist or get displaced?	Incumbent-challenger pair x day.	Lagged vehicle share and challenger route-cost edge.	Table 5	Incumbents persist, but large challenger cost edges predict share losses.
RQ4	When does vehicle status switch under stress?	Common-support stress-event days.	WETH-minus-stable vehicle share relative to a prior baseline.	Table 6	Stress rotates share away from WETH and toward stable vehicles on event days.
RQ5	How does architecture change vehicle formation?	Endpoint-pair event windows around V3.	Direct-route availability and no-direct/WETH-available cases.	Table 7	V3 expands direct-route opportunity and reduces no-direct dependence on WETH.
RQ6	How does settlement design change vehicle use?	Matched V3/V4 route cells and receipt-audited route units.	Intermediate-token transfer incidence and matched-cell V4 route use.	Table 8	V4 lowers physical intermediary transfers while vehicle-route demand persists.
RQ7	Does a vehicle create common liquidity?	Pool x vehicle x day liquidity panel.	Leave-one-out vehicle liquidity factor.	Table 9	Vehicle-linked pools share a same-vehicle liquidity component beyond market liquidity.

Table 2: Vehicle-use measurement and exact-quote scope.

Panel	Quantity	Main estimate	Companion estimate	Interpretation
<i>Panel A. Vehicle-use denominators, 2026</i>				
	WETH	44.5% conditional BridgeShare	5.3% all-route bridge share	50.0% endpoint-pair coverage
	USDC	20.6% conditional BridgeShare	2.5% all-route bridge share	25.3% endpoint-pair coverage
	USDT	23.6% conditional BridgeShare	3.0% all-route bridge share	20.7% endpoint-pair coverage
<i>Panel B. Exact-quote scope</i>				
	Exact-quote covered share	78.9		Unified leg-volume share covered by V2/Sushi V2/V3 plus Balancer weighted quote extension
	Excluded Curve+Fluid share	16.2		Unified leg-volume share not covered by exact executable-depth quotes
	Excluded / covered ratio	0.205		Maximum exact-quote scope exposure relative to covered quoteable venues
	Excluded stable-leg share	75.3		Excluded venues are stablecoin-heavy, so route-cost claims must be scoped

Notes: BridgeShare is conditional on indirect routes. All-route bridge share and endpoint-pair coverage are reported so the conditional vehicle measure is not confused with a share of all DEX volume. Exact executable-depth route-cost results are scoped to covered quoteable venues.

Table 3: Variable construction and empirical proxies.

Variable	Level	Construction	Used for
VehicleShare	vehicle token x day	USD volume of indirect routes where token v is an intermediate divided by total indirect-route USD volume that day.	RQ1, RQ2, RQ3, RQ4
RouteCostAdvantage	endpoint pair x vehicle x day x trade size	(vehicle-route exact-quote output USD - direct-route exact-quote output USD) / direct-route output USD; reported in bps as 10000 x advantage. Main core panel uses the median bps across endpoint pairs at \$10k.	RQ1, RQ3, RQ5
DirectAvailable	endpoint pair x day x trade size	Indicator that the exact-quote engine finds a direct executable route for the endpoint pair at the standard notional.	RQ1, RQ5
VehicleAvailable	endpoint pair x vehicle x day x trade size	Indicator that both legs through candidate vehicle v are exact-quote executable at the standard notional.	RQ1, RQ5
VehicleLinkedLiquidity	vehicle token x day	USD TVL in Uniswap V3 pools whose vehicle-side/base asset is candidate v, after excluding absurd pool TVL outliers.	RQ2, RQ3, RQ6
LPCConcentration	vehicle token x day	VehicleLinkedLiquidity divided by total vehicle-candidate linked liquidity across the vehicle set that day.	RQ2, RQ3
SettlementTransferMismatch	matched route unit	Indicator that the transaction receipt contains at least one ERC-20 Transfer log matching the intermediate vehicle token.	RQ6
VehicleLiquidityFactor	vehicle token x day	Leave-one-out average daily log-liquidity change among other pools linked to the same vehicle; paired with a leave-one-out market liquidity factor.	RQ7

Notes: All variables are generated from the reconstructed route and liquidity data. VehicleShare is conditional on indirect routed volume; all-route shares are used only as scope diagnostics.

Table 4: Vehicle formation: route economics, availability, and realized route choice.

Panel	Outcome	Regressor	Estimate	SE	t	p	N
<i>Panel A. Actual route choice</i>							
	Actual vehicle share	Route-cost advantage, per 100 bp	0.1462***	0.0054	27.02	<0.001	189,143
	Actual vehicle share	Vehicle route available	19.1385***	0.4836	39.58	<0.001	189,143
	Actual vehicle share	Vehicle-route depth	28.8502***	0.5780	49.92	<0.001	189,143
<i>Panel B. Future vehicle share</i>							
	Vehicle share, t+7	Route-cost advantage, per 100 bp	0.0001***	0.0000	3.10	0.002	9,380
	Vehicle share, t+7	Vehicle-route availability	0.0940***	0.0141	6.67	<0.001	9,380
	Vehicle share, t+7	LP concentration	0.1117***	0.0135	8.30	<0.001	9,380
	Vehicle share, t+30	Current vehicle share	0.3630***	0.0165	21.98	<0.001	9,265
	Vehicle share, t+30	Route-cost advantage, per 100 bp	0.0002***	0.0000	5.26	<0.001	9,265
	Vehicle share, t+30	Vehicle-route availability	0.1254***	0.0152	8.23	<0.001	9,265

Notes: Panel A uses endpoint-pair-by-date fixed effects and date-clustered standard errors. Panel B uses token and date fixed effects with date-clustered standard errors. Route-cost advantage is measured per 100 basis points.

Table 5: Liquidity provision, persistence, and challenger displacement.

Panel	Outcome / bin	Regressor / statistic	Estimate	SE	t	p	N
<i>Panel A. Liquidity-route feedback</i>							
	Vehicle share, t+7	LP concentration	0.1117***	0.0135	8.30	<0.001	9,380
	LP concentration, t+7	Current vehicle share	0.0419***	0.0098	4.29	<0.001	9,380
	Log vehicle-linked liquidity, t+7	Current vehicle share	1.1713***	0.0621	18.86	<0.001	9,380
<i>Panel B. LP stock changes</i>							
	Change in LP concentration, t to t+30	Current vehicle share	-3.4047***	1.3012	-2.62	0.009	9,265
	Change in LP concentration, t to t+30	Vehicle-route availability	3.6119**	1.4612	2.47	0.014	9,265
	Change in LP concentration, t to t+30	No-direct, vehicle-available	-6.5773**	3.1165	-2.11	0.035	9,265
	Change in log vehicle-linked liquidity, t to t+30	Current vehicle share	-0.1805***	0.0458	-3.94	<0.001	9,265
	Change in log vehicle-linked liquidity, t to t+30	Vehicle-route availability	0.2112***	0.0454	4.65	<0.001	9,265
	Change in log vehicle-linked liquidity, t to t+30	No-direct, vehicle-available	-0.4296***	0.1154	-3.72	<0.001	9,265
<i>Panel C. Challenger displacement thresholds</i>							
	Challenger <= incumbent	Incumbent share change, t+30	-2.80***		-8.16	<0.001	1,687
	0 to 25 bp	Incumbent share change, t+30	-5.19***		-3.47	<0.001	86
	25 to 100 bp	Incumbent share change, t+30	-6.72***		-5.47	<0.001	146
	100 to 250 bp	Incumbent share change, t+30	-9.50***		-7.28	<0.001	92
	>250 bp	Incumbent share change, t+30	-10.08***		-10.17	<0.001	193

Notes: Panels A and B use token and date fixed effects with date-clustered standard errors. Panel C reports bin means for challenger route-cost advantages and incumbent share changes.

Table 6: Stress rotation inside common route opportunities.

Panel	Estimate	Effect	Events	t	p	Interpretation
<i>Panel A. Same-day decomposition</i>						
	WETH share change	-1.48 pp**	20	-2.59	0.018	event-day minus prior-28-day baseline
	Stable share change	1.48 pp**	20	2.59	0.018	event-day minus prior-28-day baseline
	WETH-minus-stable change	-2.96 pp**	20	-2.59	0.018	event-day minus prior-28-day baseline
	Aggregate direct-route share change	-0.25 pp	20	-0.42	0.679	event-day minus prior-28-day baseline
	Log indirect-route volume change	0.51 log points***	20	3.2	0.005	event-day minus prior-28-day baseline
<i>Panel B. Event-time persistence</i>						
	event day	-26.96***	20	-3.06	0.006	WETH-minus-stable gap
	post 1 to 7	-11.16*	20	-1.87	0.077	WETH-minus-stable gap
	post 8 to 30	-9.48**	20	-2.13	0.046	WETH-minus-stable gap
	pre -14 to -1	-5.63**	20	-2.63	0.016	WETH-minus-stable gap
<i>Panel C. Threshold and overlap sensitivity</i>						
	all events, threshold 6%	-3.12 pp***	91	-3.44	<0.001	negative share 67.0%
	all events, threshold 8%	-2.95 pp***	49	-3.41	0.001	negative share 65.3%
	all events, threshold 10%	-3.74 pp***	26	-3.7	0.001	negative share 65.4%
	all events, threshold 12%	-3.18 pp**	19	-2.68	0.015	negative share 63.2%
	top 20, threshold 8%	-3.09 pp**	18	-2.48	0.024	negative share 61.1%
	non-overlap, threshold 8%	-3.01 pp**	33	-2.45	0.020	negative share 57.6%

Notes: Event-day effects are measured relative to a prior 28-day baseline. The event-time rows show that pre-movement exists, so the stress result should be written as a same-day common-support rotation, not a clean surprise-shock causal design.

Table 7: Architecture and direct-route opportunity.

Panel	Outcome	Sample	Effect	SE	t	p	N
<i>Panel A. Main V3 opportunity test</i>							
	No-direct, WETH-available	All matched pairs	- 25.81***		-7.62	<0.001	1,822
<i>Panel B. Dose response by pre-V3 direct availability</i>							
	Direct-route availability	Q1 weakest	56.42***	10.03	5.63	<0.001	1,122
	No-direct, WETH-available	Q1 weakest	- 33.03***	12.22	-2.70	0.007	1,122
	Direct-route availability	Q2	51.36***	14.83	3.46	<0.001	1,434
	No-direct, WETH-available	Q2	- 47.51***	8.40	-5.66	<0.001	1,434
	Direct-route availability	Q4 strongest	3.28*	1.98	1.66	0.099	8,914
	No-direct, WETH-available	Q4 strongest	-1.91	1.98	-0.96	0.336	8,914

Notes: Effects are percentage points unless otherwise noted. The table is scoped to route opportunity around V3 and should not be read as a broad causal launch effect for every V3 outcome.

Table 8: Settlement design, physical transfer incidence, and matched-cell route use.

Panel	Sample / outcome	Cells / N	V3 / regressor	V4	Difference / estimate	p / diagnostic
<i>Panel A. Physical intermediary-token transfer incidence</i>						
	All	500	100.0%	81.4%	-18.6 pp***	<0.001
	Route size: Small	104	100.0%	60.6%	-39.4 pp***	<0.001
	Route size: Medium	73	100.0%	87.7%	-12.3 pp***	0.002
	Route size: Large	94	100.0%	96.8%	-3.2 pp*	0.083
<i>Panel B. Matched-sample balance and receipt diagnostics</i>						
	Uniswap V3	500	\$6,136		53.05	100.0
	Uniswap V4	500		\$3,542	71.05	81.4
	V4 minus V3 within cell	500			log route diff=-0.692	t = -7.58, p < 0.001
<i>Panel C. Matched-cell route-use persistence</i>						
	Log V4 route count	16,218	Log V3 route count		0.3674 (t=28.33)***	<0.001
	Log V4 route volume	16,218	Log V3 route volume		0.5819 (t=83.13)***	<0.001

Notes: Panel A compares matched V3 and V4 route units. Panel B reports matched-sample route-size and receipt-log diagnostics; numeric entries in the last column are diagnostic percentages or test statistics, not p-values unless labelled. Panel C shows that V4 route demand persists in matched endpoint-vehicle-week cells.

Table 9: Common liquidity across pools linked to the same vehicle.

Panel	Sample	Regressor	Estimate	SE	t	p	N
<i>Panel A. Baseline common-liquidity regressions</i>							
	Full sample	Market liquidity factor	0.7063***	0.0573	12.32	<0.001	685,685
	Full sample	Vehicle liquidity factor	0.1626***	0.0431	3.77	<0.001	685,685
	Stress interaction	Market liquidity factor	0.6987***	0.0605	11.55	<0.001	685,685
	Stress interaction	Vehicle liquidity factor	0.1586***	0.0438	3.62	<0.001	685,685
	Stress interaction	Vehicle factor x stress	0.1064**	0.0491	2.17	0.030	685,685
<i>Panel B. Heterogeneity and top-pool exclusion</i>							
	High average VehicleShare vehicles	Vehicle liquidity factor	0.2174***	0.0464	4.69	<0.001	664,218
	Low average VehicleShare vehicles	Vehicle liquidity factor	-0.0921	0.0903	-1.02	0.307	21,467
	Excluding top 1% mean-liquidity pools	Vehicle liquidity factor	0.1670***	0.0436	3.83	<0.001	679,228

Notes: Regressions include pool-vehicle fixed effects and date-clustered standard errors. The vehicle factor is leave-one-out across other pools linked to the same vehicle.

Table 10: Specification registry for paper-facing tests.

Test	Unit	Sample	Outcome	Treatment / regressor	FE / SE	Main estimate	Interpretation
P1 availability/ direct	endpoint- pair x day x trade size	V2/Sushi V2/V3 exact quoteable venues	route availabil- ity and WETH advantage	WETH vehicle route	endpoint- pair-day aggrega- tion	9,584 no-direct rows; thin-direct 142.65- 349.28 bp	availability and thin-direct execution protection
P2 liquidity- route feedback	token x day	vehicle candidates	future Bridge- Share; future LP concentra- tion/liquidity	LP concen- tration; current Bridge- Share	token/date FE; date- clustered SE	two-way coefficients positive; p<0.001	reduced-form liquidity-route feedback
P3 impact stress	event x common endpoint- pair set	WETH downside event days	WETH- minus- stable Bridge- Share	event-day WETH downside stress	event- level t-tests	-2.96 pp, p=0.018	same-day rotation away from WETH toward stable vehicles
P4a V3 opportu- nity	endpoint- pair x month	balanced V3 launch- window pairs	no- direct/WETH- available indicator	post-V3	pair FE; pair- clustered SE	-25.81 pp, p<0.001; pretrend p=0.922	direct-route opportunity expansion
P4b V4 settle- ment	matched route unit; token x week	matched V3/V4 cells; LP panel around launch	transfer incidence; LP liquidity response	V4 route unit; post x netting exposure	matched- cell tests; to- ken/week FE	V4 gap -18.6 pp; log LP beta 2.132	settlement netting lowers movement and predicts LP supply response

Notes: This registry fixes the paper-facing unit, sample, outcome, treatment or regressor, inference convention, headline estimate, and bounded interpretation for each main test.